



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	ORGANIC CHEMISTRY 1				
Type of Course	MAJOR /MINOR				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basics of organic chemistry-Functional groups, Homologous series, Nomenclature and isomerism				
Course Summary	This course explores basics of organic chemistry reaction mechanism, Reactions and mechanism of important functional groups and stereochemistry				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To understand the basics of Organic chemistry	U	C	Test /Seminar
CO2	To understand the basic concepts of reaction mechanisms	U	p	Discussion/ Assignment
CO3	To recognize the various types of organic reactions and reaction intermediates	An	P	Quizzes/Test
CO4	To realise the importance of stereoisomerism, optical activity and chirality	Ap	P	Discussion/Seminar /Assignment
CO5	To enable the students to improvise Molecular models	Ap	P	Assignment/Test

CO6	To empower students in various separation and purification techniques	Ap	P	Lab work/Viva
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction		12	26
	1	IUPAC Nomenclature of multifunctional acyclic and cyclic compounds. Structural isomerism.	2	
	2	Hybridization and bonding in organic compounds (methane, ethane, ethylene and acetylene)	2	
	3	Localised and delocalised bonding. Hydrogen bonding, effect of hydrogen bonding on physical and chemical properties of compounds	1	
	4	Organic acids and bases	2	
	5	Basics of MO theory as applied to organic molecules -Ethylene and Buta-1,3-diene.	3	
	6	Aromaticity-Huckel's rule for aromaticity (Benzenoid compounds)	2	
II	Organic reaction mechanisms		12	26
	7	Types of bond fission-Homolytic and Heterolytic fission	1	
	8	Arrow formalism used in reaction schemes.	1	
	9	Electrophiles and Nucleophiles	1	
	10	Electron displacement Effects: Inductive effect and Field effect, Steric effect- Acidity and basicity of organic compounds based on Field effect and steric effect.	2	
	11	Electromeric effect, Mesomeric effect	2	
	12	Hyperconjugation- Stability of alkenes.	1	
	13	Reactive intermediates: Structure, formation and stability of carbocations, carbanions, free radicals, carbenes and nitrenes.	3	
	14	Pericyclic reactions and its classifications	1	

III	Stereochemistry-I		14	30
	15	Stereoisomerism: Conformational isomerism and configurational isomerism. Representation of stereostructures of organic molecules using Flying wedge, Fischer, Sawhorse and Newmann projections.	3	
	16	Inter conversion of different projections of L-tartaric acid and 3-chloro-2-butanol.	3	
	17	Conformational Isomerism – Conformational analysis of Ethane, n- butane and cyclohexane with PE diagram.	3	
	18	Conformation of mono substituted cyclohexanes. Relative stability of conformations.	2	
	19	Configurational isomerism: Geometrical isomerism in alkenes, cycloalkanes and oximes. Cis-trans, Syn-Anti and E-Z notations, sequence rule.	3	
IV	Purification and Characterization Techniques		7	16
	20	Distillation- Simple, fractional, steam and vacuum distillations	2	
	21	Recrystallisation, sublimation, solvent extraction.	2	
	22	Chromatography, stationary phase, mobile phase, R _f values, - TLC, Column chromatography, HPLC and GC (basic concepts only).	3	
V	Practicals		30	
	1.	Introduction to organic lab	4	
	2	1. Distillation of Aniline, 2. Distillation of Limonene (from orange peels) 3. Purification of organic compounds by crystallization using the following solvents: a. Water b. Alcohol 4. Sublimation of a dicarboxylic acid/Naphthalene 5. Molecular model construction and conformation of ethane 6. Molecular model construction of Ethylene or Acetylene 7. Molecular model construction of acetaldehyde and Cyclohexane.	20	
	3	Open ended	6	

References

1. Morrison, R. N. & Boyd, R. N., Organic Chemistry, Dorling Kindersley (India) Pvt. Ltd. (Pearson Education).
2. Bhal and Bhal, Advanced Organic Chemistry, 2nd Edition, S. Chand Publisher, 2012.
3. Kalsi, P. S., Stereochemistry Conformation and Mechanism; 8thEdn, New Age International, 2015
4. I. L. Finar, *Organic Chemistry*, Vol. I, 5th Edn., Pearson Education, New Delhi, 2013.
5. M. K. Jain, S. C. Sharma, *Modern Organic Chemistry*, 3rd Edn., Vishal Publishing Company Co., 2010.
6. K. S. Tewari, N. K. Vishnoi, S. N. Mehrotra, *A Textbook of Organic Chemistry*, 2nd Edn., Vikas Publishing House, New Delhi, 2004.
7. B. S. Furniss, A. J. Hannaford, P. W. G. Smith, A. R. Tatchell, *Vogel's Textbook of Practical Organic Chemistry*, 5th Edn., Pearson Education, Noida, 2014.
8. F. G. Mann, B. C. Saunders, *Practical Organic Chemistry*, 4th Edn., Pearson Education, Noida, 2011.
9. Arthur I. Vogel, *Elementary Practical Organic Chemistry- Small Scale Preparations*, 2nd Edn., Pearson Education, Noida, 2013
10. An Improved Method for the Extraction and Thin-Layer W Chromatography of Chlorophyll a and b from Spinach Hao T. Quach, Robert L. Steeper, and G. William Griffin, J Chem Edn, 2004, 81, 385
11. Chemistry for Pharmacy Students: General, Organic and Natural Product Chemistry, S D Sarkar and L Nahar, John Wiley and sons, Ltd.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1						3				1	1	1
CO 2	2						2				2		1
CO 3	3						2				2		1
CO 4				2	2		2				2		1
CO 5	2						2		1	1	1	1	1
CO 6			3			2	2		1		2	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/viva/seminar	Practical skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2		✓		✓
CO 3	✓			✓
CO 4		✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	✓